

Rupam Ghosal

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Professional Summary

PhD Research Scholar in Artificial Intelligence, Machine Learning, Physics and Robotics with deep expertise in autonomous systems, predictive modeling, and sustainable AI design. Proven track record in developing production-ready AI/ML models, scalable algorithms, and robotics automation for high-impact domains including healthcare, aerospace, and intelligent control systems.

Skilled in computer vision, deep learning, reinforcement learning, natural language processing (NLP), and IoT integration, with a strong emphasis on energy-efficient and ethically responsible AI solutions. Adept at bridging theoretical physics with applied engineering to deliver innovative, reproducible, and automation-driven technologies. Recognized for architecting intelligent systems that optimize performance, enhance decision-making, and drive sustainability. Actively seeking to contribute to cutting-edge AI innovation within globally leading technology companies.

Experience

PhD Research Scholar – AI & Physics

Birla Institute of Technology, Mesra | Jan 2025 – Present

- i. Built offline LLM agents using LangChain, Python, Miniforge for remote-access environments
- ii. Developed SLAM-based robots and RL controllers for autonomous navigation and energy optimization
- iii. Integrated AI frameworks with robotics hardware to deliver scalable prototypes

Teaching Associate – Robotics & Physics

D Y Patil International University, Pune | Aug 2023 – Sep 2024

- I. Delivered lectures on- BioPhysics, AI/ML, automation, and robotics; mentored student projects using Python, Fusion 360, AutoCAD
- II. Led hands-on labs on DTFW [Design Thinking and Fabrication] sensor integration and control systems; and Biophysics btech practicals.

STEM Trainer – Robotics & IoT

Tinkerworks (GSL), Ranchi | Jun 2023 – Aug 2023

- I. Conducted workshops on Arduino, IoT, and AI-powered automation
- II. Guided student teams in building drones, RC cars, and smart alarm systems

Education

- ❖ PhD in Physics (*AI & Healthcare Focus*)
Birla Institute of Technology, Mesra | Jan 2025 – Present
- ❖ PhD Coursework & Research in Physics
D Y Patil International University, Pune | Sep 2023 – Sep 2024
- ❖ Integrated Master of Science (IMSC Physics) – CGPA 7.0
Birla Institute of Technology, Mesra | Aug 2018 – Jun 2023

Projects

1. **Offline ChatGPT for Remote Access**
Built a local LLM chatbot with LangChain, Miniforge, and Python; achieved sub-200 ms response times on edge devices without internet
2. **AI-Powered Healthcare Device**
Developed a CNN-based diagnostic tool for early disease detection; improved classification accuracy by 12% over existing benchmarks
3. **AI-Driven Autonomous Research Assistant**
Engineered a hybrid system combining LLMs, episodic memory modules, and voice UI to automate lab workflows, cutting routine experiment time by 40%
4. **Aerospace Materials Innovation**
Synthesized AI-doped MoS₂ solid lubricants, reducing friction by 25% under vacuum conditions
Engineered diamond-coated titanium thin films for bearings, boosting wear resistance by 30%

Technical Skills

- **Core AI/ML:** Deep Neural Networks · Large Language Models · Reinforcement Learning · Computer Vision · NLP · Generative AI · SLAM
- **Programming & Tools:** Python · Git · Miniforge · Pandas · NumPy · LangChain · MATLAB · Arduino IDE
- **Development:** React · Full-Stack AI Integration · Conversational & Offline AI Agents
- **Engineering & Research:** Data Analysis · Predictive Modeling · Instrument Calibration · Robotics Automation · IoT Integration
- **Design & Office:** AutoCAD · Fusion 360 · Microsoft Office · Adobe Creative Suite

Languages

English (Fluent) · Hindi (Native) · Bengali (Native) · Marathi (Intermediate) · Japanese (Beginner).